

Baseline Models Report

Machine Learning for Gunshot Residue Classification: Accuracy, Interpretability, and Failure Analysis

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Executive Summary

This report presents findings from initial baseline modeling across three different modalities (logistic regression, XGBoost, neural network).

A baseline logistic regression model was fit for the linear modeling track using an engineered feature set derived from elemental compositions. The model was trained on a binary classification task (GSR vs. Non-GSR) after excluding ambiguous single-element particles., with a group-aware train-test split applied using `stud_id` to prevent leakage. `StandardScaler` was used to normalize features prior to fitting. Initial performance was near-perfect, with ROC-AUC approaching 1.0 and accuracy exceeding 99% on the held-out test set. To address concerns around feature dominance, key elemental predictors (Pb, Ba, Sb) were removed in subsequent iterations; however,

performance remained extremely high, indicating that the remaining features still provide strong discriminative signal.

Coefficient analysis confirmed that the model relies on co-occurrence patterns of elements rather than any single predictor, with misclassifications primarily occurring in barium-containing particles, consistent with prior EDA findings. Overall, results indicate that the binary GSR classification task is highly separable using a linear decision boundary when multi-element signatures are present. As such, the more meaningful evaluation lies in model behavior on ambiguous particles, where uncertainty reflects inherent data ambiguity. The logistic regression model serves as an interpretable baseline for comparison with more complex approaches.

XGBoost was selected as the tree model based on exploratory findings of complex non-linear relationships across elemental composition as well as the sparsity of the data.

Three XGBoost baseline versions were explored for this report, each version trimming the number of engineered features based on initial model performance analysis.

Baseline version 1 has the most features with a total of 41, while version 3 had the fewest with 37. All models consisted of the same 27 raw elemental compositions, then ranged between 10, 13, and 14 engineered features.

Ultimately, version 1 was dropped from the model exploration due to suspected feature leakage or label proxy, while versions 2 and 3 removed the problematic feature. The important distinction between these remaining versions is that version 3 removed an additional 3 engineered features to assess potential redundancy, feature leakage, or label proxies. Appendix E contains in-depth feature analysis for the potential leakage

removed from versions 2 and 3, as well as further details regarding engineered features that exist in version 2 but not in version 3.

Both version 2 and version 3 baseline models performed with incredible precision, but version 2 had a slight advantage with false-positive rate. It is worth noting that the removal of the specific engineered features from version 3 seem to have allowed it to distribute its split importance across more raw elements than version 2, while only suffering slightly in performance. This allows the tree model to work from raw chemical measurements rather than what *may* be shortcut aggregates causing overperformance.

A baseline multi layer perceptron was fit for the neural network track and trained on a 56-feature representation consisting of CLR-transformed elemental compositions, binary presence indicators, and two high-signal engineered features. After initial fitting, the GSR count feature was dropped since near-perfect performance (ROC-AUC 1.0000) indicated significant target leakage, effectively acting as a proxy for the label. After removing it, the baseline MLP still achieved 99.9% accuracy, ROC-AUC of 1.0, and a false positive rate of 0.07% on the held-out test set, with consistent results across 3-fold group-aware cross-validation. We believe that this reflects the fact that the binary GSR vs. Non-GSR classification task is well-separated in the feature space, confirming that the feature engineering pipeline captures strong discriminative signal.

The more revealing finding came from scoring the 506,682 ambiguous particles (single-element classes Pb, Ba, Sb, Sr) that were excluded from training. Here, the model classified 97–99% of them as Non-GSR with high confidence, and only 1–2% of Pb-only and Sb-only particles received GSR predictions. This aligns with forensic domain

knowledge that single-element particles lack the multi-element signature required for reliable GSR identification, and validates the original decision to label these particles as ambiguous. Next steps for the neural network track include exploring TabNet (Arik & Pfister 2021) for interpretable attention-based feature selection, threshold optimization for forensic applications, and cross-dataset validation against NIST ground-truth data. We also plan to apply MC Dropout uncertainty estimation to quantify prediction confidence for ambiguous single-element particles where the model's uncertainty may reflect genuine aleatoric uncertainty in the underlying forensic data. While other methods exist that explicitly address uncertainty, e.g. Bayes via Backprop (Blundell, Cornebise, Kavukcuoglu & Wierstra 2015) or Evidential Regression (Sensoy, Kaplan, & Kandemir 2018), these methods require significant implementation time and we reserve them for possible future exploration but exclude them from the current set up.

1. Background and Research Question

The identification of GSR particles is a cornerstone of forensic investigations involving firearm discharge. When a firearm is fired, microscopic particles containing characteristic elements such as lead (Pb), barium (Ba), and antimony (Sb) are deposited on nearby surfaces and can serve as physical evidence in criminal proceedings. Traditional GSR analysis relies on scanning electron microscopy with energy dispersive X-ray spectroscopy (SEM/EDS), a process that depends heavily on expert judgment and is susceptible to both false positives from chemically similar environmental particles and false negatives from atypical GSR compositions.

Redefined Research Question

How accurately and reliably can machine learning models distinguish true gunshot residue particles from chemically similar non-GSR particles, and what are the limits of these models when applied to ambiguous single element particles? We further investigate how classification performance varies across model types (linear, tree-based, neural network), which elemental features drive predictions, and under what conditions models fail, particularly regarding false positives where environmental particles are incorrectly labeled as GSR.

2. Redefined Hypothesis & Prediction

We hypothesize that machine learning models can effectively distinguish true GSR particles from environmental confounders, when the classic multi-element signatures (Pb-Ba-Sb combinations) are present, achieving near-perfect classification on the binary task, with non-linear models (e.g., gradient-boosted trees, neural networks) slightly outperforming linear models (e.g., logistic regression). We further hypothesize that single element particles will lead to substantial uncertainty in model classifications with degraded performance compared to multi element particles. This reflects genuine forensic uncertainty, or aleatoric uncertainty.

We predict that feature importance analyses (e.g., SHAP, TabNet attention mask, regression coefficients) will identify lead, barium, and antimony, along with their ratios, as the most influential predictors, and that the co-occurrence of multiple GSR elements within a single particle will be more predictive than any individual element's concentration. We further predict that misclassifications will concentrate among barium-heavy Non-GSR particles (BaAl, BaCaSi) and single-element ambiguous particles,

highlighting the boundary where automated classification should defer to human expert judgment.

3. Methods

The following sections give an overview of the fitting and evaluation process for each baseline model as well as adjustments made from the modeling plans outlined in the feature engineering and preprocessing report.

3.1 Logistic Regression

Logistic regression was implemented as a baseline linear classification model to establish a benchmark for comparison with more complex modeling approaches. The model was designed to predict binary class labels (GSR vs. Non-GSR) using engineered elemental features derived from SEM/EDS particle data.

Prior to modeling, the dataset was restricted to particles labeled as either GSR or Non-GSR, excluding ambiguous single-element classes to ensure a well-defined binary classification task. Predictor variables were limited to numeric elemental features, while identifier and target-related fields were excluded to prevent data leakage.

During feature exploration, several elemental predictors, specifically lead (Pb), barium (Ba), and antimony (Sb), were identified as highly dominant features that nearly perfectly separated the target classes. To evaluate the impact of these predictors on model behavior, two modeling configurations were developed:

- **Original Model:** Included all numeric elemental features

- **Reduced Model:** Excluded Pb, Ba, and Sb to assess model performance without dominant signals

To mitigate data leakage at the sample level, a group-based train-test split was applied using stub id as the grouping variable. This ensured that all particles from a given stub were contained entirely within either the training or testing dataset, preventing the model from learning stub-specific patterns.

Both models were implemented using a standardized pipeline consisting of median imputation, feature scaling via StandardScaler, and L2-regularized logistic regression. Identical preprocessing and hyperparameters were used across both configurations to ensure that performance differences were attributable solely to the inclusion or exclusion of dominant elemental predictors.

Model performance was evaluated using confusion matrices, classification reports, receiver operating characteristics (ROC) curves, and Precision-Recall (PR) curves, with additional emphasis on interpretability through analysis of model coefficients.

3.2 XGBoost

For XGBoost, splitting the data followed the same approach that will be described in section 3.3 for the neural network: a two-stage GroupShuffleSplit using stub_id as the grouping variable to produce a 60/20/20 train/validation/test split with zero stub overlap between any pair of partitions to prevent training leakage. The validation set was used exclusively for early stopping, while the test set was held out for final evaluation. No feature scaling or normalization was applied. XGBoost operates on binary split

thresholds and is invariant to feature magnitude, unlike logistic regression and neural networks which require standardized inputs.

Both tree model versions used the following hyperparameters:

- threshold=0.5
- n_estimators=500
- max_depth=6
- learning_rate=0.1
- min_child_weight=5
- subsample=0.8
- colsample_bytree=0.8
- reg_alpha=0.1
- reg_lambda=1.0
- early_stopping_rounds=30
- eval_metric='aucpr'
- tree_method='hist'

These represent moderate defaults intended to establish a performance baseline before formal hyperparameter tuning.

Three-fold cross-validation was performed using StratifiedGroupKFold on the combined training and validation set, representing 80% of the data and maintaining stub-level integrity within each fold. The test set was held out and entirely untouched during cross-validation. The CV results suggest that both models generalize stably across different stub groupings.

XGBoost's core assumptions are:

1. The binary labels are correct. The expert-assigned GSR/Non-GSR labels are considered ground truth.
2. The selected features are informative for the target.
3. The training data is representative of the deployment distribution strategy that ensures generalization to unseen physical samples.
4. The model complexity is appropriate for the signal in the data, which is controlled through regularization and early stopping.
5. Timelessness. The elemental compositions are assumed to be a snapshot of the particle in time, and it is also assumed that these compositions and the particles' GSR status doesn't vary by case or time period. This is a difficult assumption to support but is necessary for the scope of this research.
6. Uniformity in collection methods and particle processing. It is assumed that across the stubs, the samples were collected in the same manner and processed using the same quantification methods. We do not have access to the details of data collection methods so these assumptions are necessary. Likewise, it is assumed that the software capturing the particle data used the same configurations and didn't falter or mishap during collection.

Unlike linear models, XGBoost does not assume linearity, feature independence, or normally distributed residuals. These non-assumptions are supported by various exploratory methods in the EDA report including the lack of PCA's explained variance in comparison to the well-defined structures from UMAP.

3.3 Neural Net (MLP)

As our baseline, we fit a simple fully connected multi layer perceptron (MLP) on a 57-feature representation consisting of 27 CLR-transformed elemental compositions, 27 binary presence indicators, and 3 engineered features. The full model consisted of 2 hidden layers, 128 units, ReLU, dropout 0.3., Adam optimizer, BCEWithLogist loss, learning rate 1e-3, resulting in a total of 23,937 parameters. This initial architecture is kept simple and untuned on purpose. We split the data into 60% train, 20% validation, and 20% test using a two stage GroupShuffleSplit on stub id to ensure that all particles from the same stub appear exclusively in the same split to prevent leakage of stub specific patterns that the model could use to learn shortcuts. StandardScaler was fit on the training data only and applied to validation and test sets. Early stopping with patience set to 5 epochs was used to monitor validation loss, and the best model was restored after training for 30 epochs.

We performed 3 fold cross validation using GroupKFold on the train and validation portions of the data, leaving the test set entirely untouched, with a fresh model trained for each fold and StandardScaler fit independently per fold. This was done to control overfitting and to verify that performance is stable across folds.

There are several core assumptions for neural networks that we addressed in our preprocessing and feature engineering. First, inputs must be real valued and unconstrained, which we satisfied by applying the CLR transform. Second, gradient based optimization required normalized inputs, which is why we applied StandardScaler to our data to center features at $\mu = 0$ and unit variance. Lastly, the loss function used

(binary cross entropy loss) assumes well calibrated probabilities. While we did not explicitly test this assumption for the baseline model, the probabilities on ambiguous particles, as discussed in more depth in the results section, appear reasonable with most GSR predictions near zero. Calibration analysis is planned for the next phase.

4. Results

4.1 Logistic Regression

The logistic regression analysis revealed a clear distinction in model performance depending on whether dominant elemental predictors were included.

The original model, which retained all elemental features including Pb, Ba, and Sb, achieved near-perfect classification performance on the held-out test set. ROC-AUC approached 1.0, and Precision-Recall AUC was similarly near 1.0, indicating almost complete separation between GSR and Non-GSR particles. However, inspection of model coefficients confirmed that this performance was driven overwhelmingly by a small number of highly predictive features, suggesting that the model was effectively reconstructing the target label using deterministic signals.

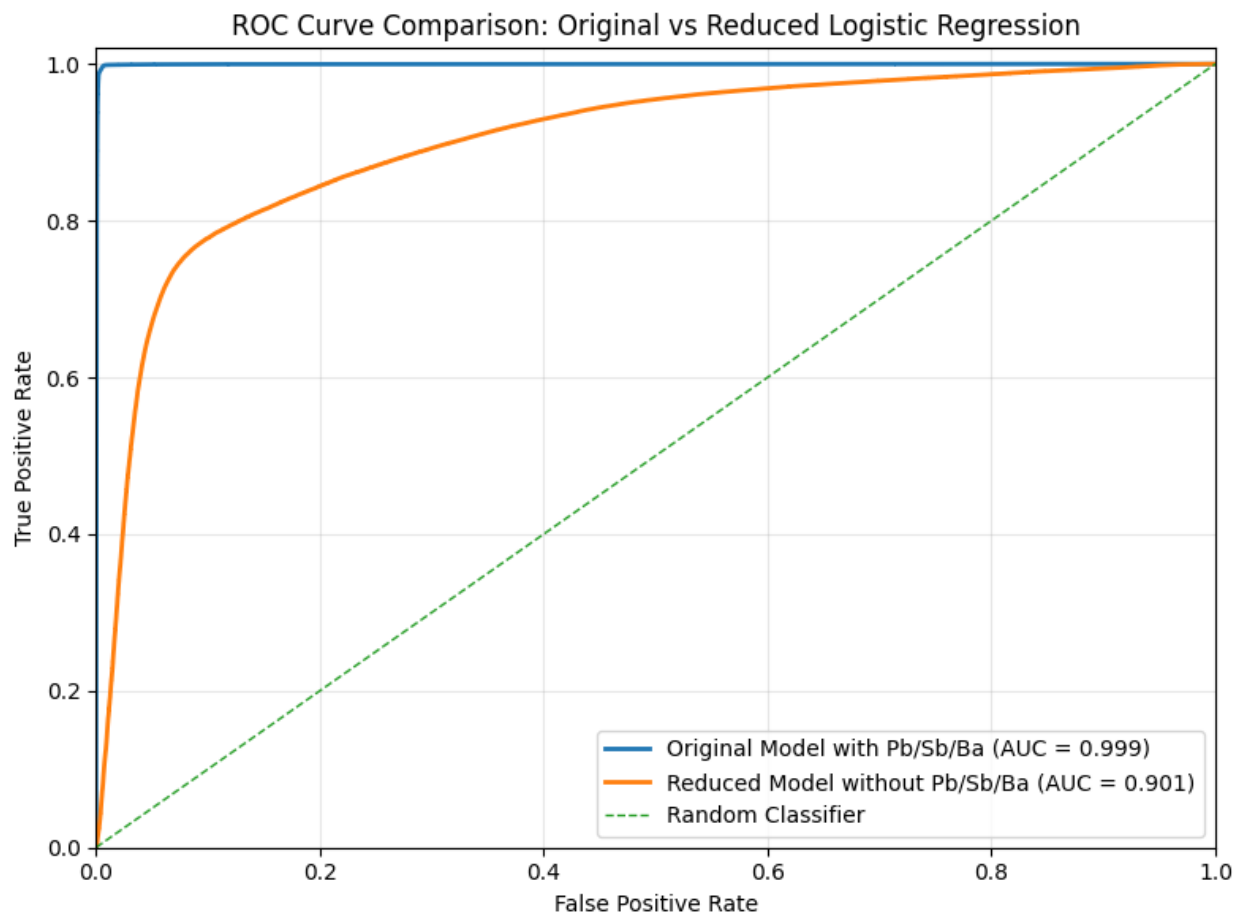


Figure 1: ROC Curve Comparison (Original vs. Reduced Models)

To evaluate the robustness of the model beyond these dominant predictors, a reduced model was developed with Pb, Ba, and Sb removed from the feature set. Following their removal, model performance decreased to a more realistic level, with ROC-AUC of approximately 0.90 and PR-AUC showing a corresponding decline. While performance remained strong, the reduction demonstrates that these elements account for a substantial portion of the separability in the dataset.

Analysis of the confusion matrix for the reduced model shows that the majority of observations are still correctly classified, though with an increase in both false positives

and false negatives relative to the original model. Misclassifications remain concentrated in particle classes containing barium, reinforcing findings from exploratory data analysis that Ba contributes to overlap between GSR and environmentally derived particles. Precision-Recall analysis further highlights differences between the models. The original model maintains extremely high precision across nearly all recall levels, reflecting its near-deterministic behavior. In contrast, the reduced model exhibits a more gradual precision-recall tradeoff, indicating that predictions are based on broader multi-element patterns rather than a small set of dominant features.

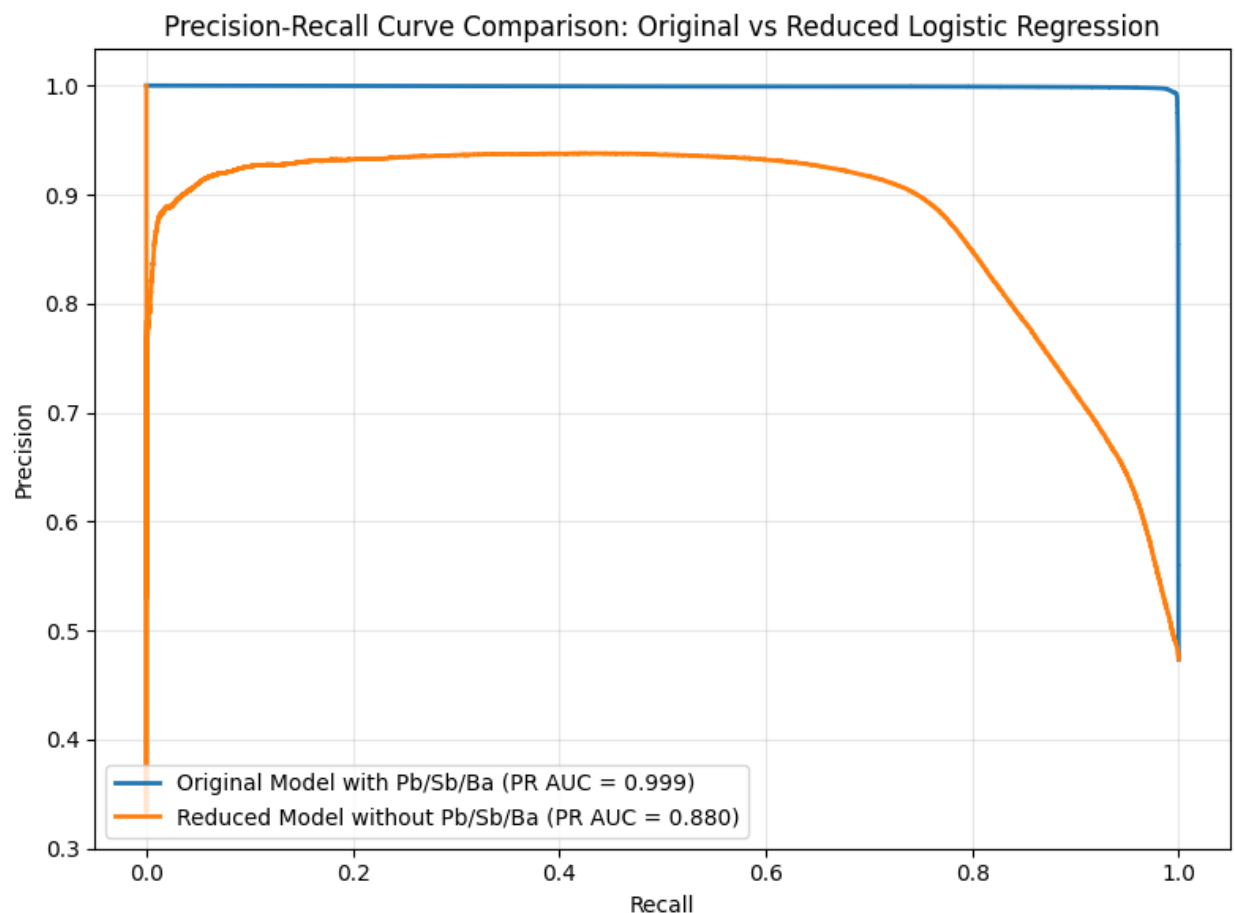


Figure 2: Precision–Recall Curve Comparison

Feature importance analysis, based on model coefficients, supports this interpretation. In the original model, Pb, Sb, and Ba exhibit disproportionately large coefficient magnitudes, confirming their role as primary drivers of classification. In the reduced model, importance is distributed across a wider set of elements, including tungsten (W), iron (Fe), zinc (Zn), and sulfur (S), suggesting that the model is capturing more generalizable compositional relationships.

Table 1: Top Logistic Regression Coefficients - Original Model

Feature	Coefficient	Abs_coef
Pb	23.045239	23.045239
Sb	12.290677	12.290677
Fe	-4.656458	4.656458
Ba	-4.535576	4.535576
Cu	-4.002127	4.002127
O	-3.970287	3.970287
Hg	-3.949162	3.949162
Zn	-3.385957	3.385957
Gd	-3.117844	3.117844
S	-1.627697	1.627697
Ga	-1.285048	1.285048
Si	-1.203463	1.203463
Ti	-1.202684	1.202684
Al	-1.113368	1.113368
Cl	-0.898865	0.898865

Table 2: Top Logistic Regression Coefficients - Reduced Model

Feature	Coefficient	Abs_coef
W	-2.641683	2.641683
Fe	-1.818156	1.818156
Zn	-1.647419	1.647419
S	-1.358866	1.358866
O	-0.887540	0.887540
Cu	-0.834088	0.834088
Gd	-0.804775	0.804775
Ga	-0.793823	0.793823
Al	-0.786109	0.786109
Si	-0.585391	0.585391
Mg	-0.545554	0.545554
Ti	-0.539042	0.539042
Cl	-0.530813	0.530813
Ca	-0.509759	0.509759
Au	-0.509397	0.509397

Overall, the results indicate that while the binary GSR classification task is highly separable when dominant elemental signatures are present, meaningful predictive structure remains even after their removal. The reduced model therefore provides a more realistic assessment of model performance and generalization capability.

4.2 XGBoost

The results for XGBoost baseline model version 2 (v2) and version 3 (v3) are both very similar, performing with outstanding precision and favorable false positive rates. A comparison of the results may shed light as to whether the 3 additional engineered features that exist in v2 are redundant, or if they truly carry significant signal that is not captured in the trimmed v3. The difference in performance may not be substantial enough to offset potential benefits of the feature distribution that v3 offers. This consideration will be evaluated more closely when feature importance is analyzed.

The table below shows the summary statistics of the split data for v2. Note that the only distinction between the two versions is the additional 3 engineered features that were kept in v2 but removed from v3, so the table for v3's split data summary statistics is not displayed. The only number that would differ is "Total features", which is 37.

Table 3: XGB Baseline Model (v2) - Split Data Summary Stats

Total observations : 2,294,985						
Total features : 40						
Total unique stubs : 3,786						
Split	Observations	% of Total	GSR (target=1)		GSR %	Unique Stubs
Train	1444147	62.9%	717835	49.71%		2271
Val	407614	17.8%	151162	37.08%		757
Test	443224	19.3%	209949	47.37%		758

The learning curves for both versions are plotted below using AUC-PR as the evaluation metric for precision across the training and validation sets. Both model versions show the validation set shooting up quickly at the very start. However, there are key differences that reveal important insights. Firstly, the training set for v2 doesn't surpass the validation set until just shy of 50 boosting rounds, whereas v3 does so right around

10 rounds. While v2 has slower convergence, it also has a smoother gradient that outperforms the early ceiling hit by v3. The engineered features kept by v2 focus on the Pb+Sb additive relationship on a logarithmic scale as well as the Pb*Sb multiplicative ratio over the total non-barium particle mass. We can deduce that these features likely provide a richer optimization surface, resulting in the smoother gradient of signal for model v2 to exploit.

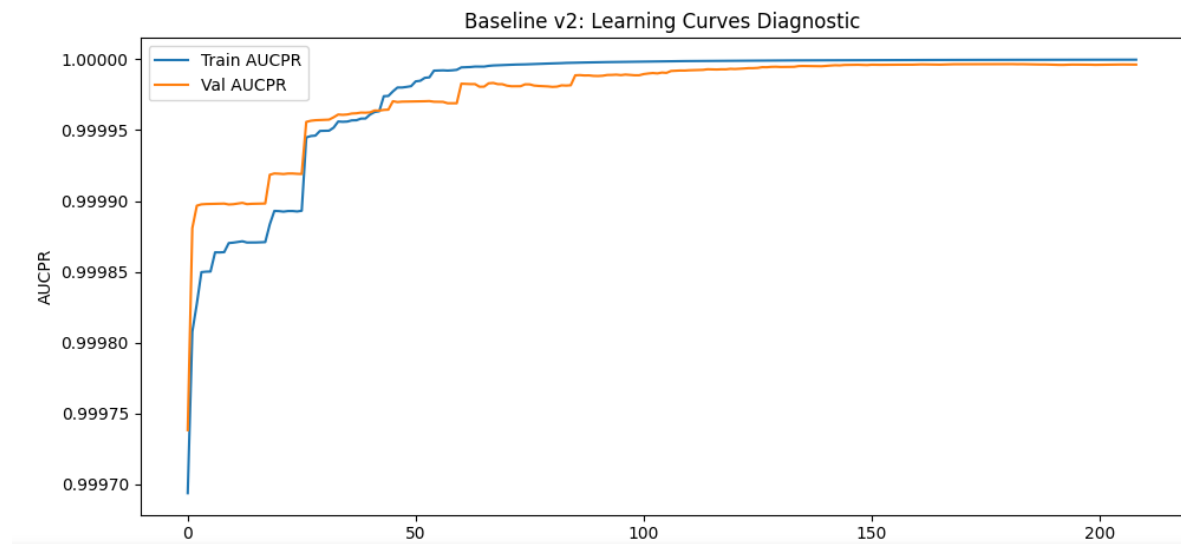


Figure 3: AUC-PR Learning Curves for Baseline Model (v2)

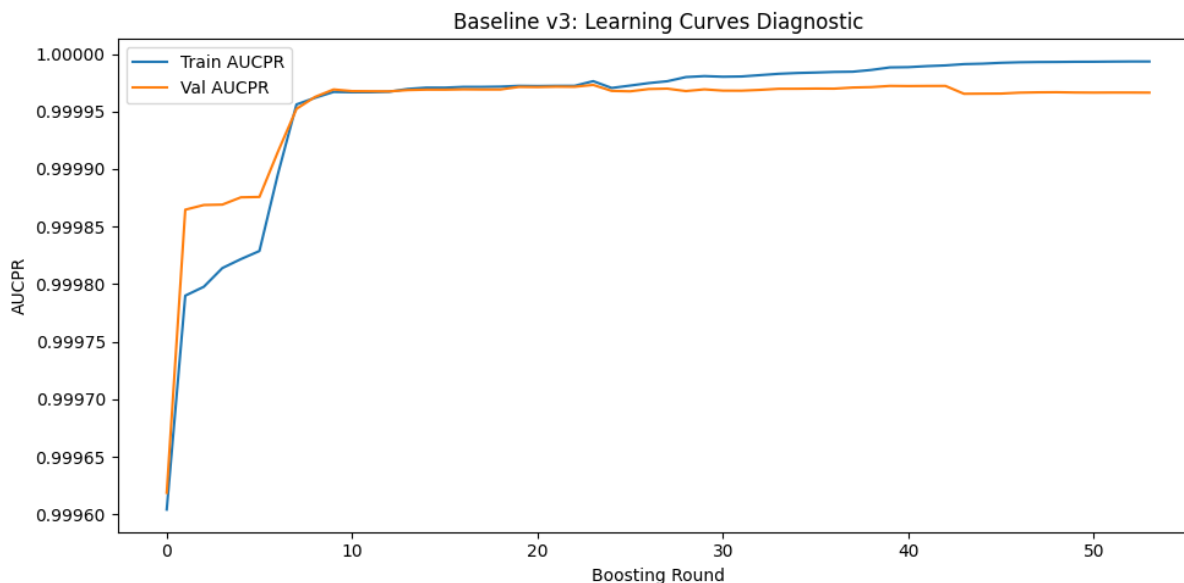


Figure 4: AUC-PR Learning Curves for Baseline Model (v3)

In contrast, the fast start but early ceiling of v3 can be interpreted as raw elemental splits on Pb and Sb that resolve the classification quickly but then run out of easily learnable signal as the barium boundary draws near, especially near round 30 when divergence becomes visible. It's likely that the barium-heavy particles don't generalize well to the validation for v3, requiring early stopping just after round 50 as the model likely recognizes that the remaining barium-heavy particles cannot be resolved with additional boosting, at least not without some tuning.

These are not arbitrary observations. The feature importance rankings in the tables below provide additional support for these inferences. The left table shows ~97% of v2 feature importance being split between the Pb and Sb engineered features previously described. The next highest contributor comes in at just 1%. There are no raw elements in the top 5. In fact, there are only 3 raw elements in the top 10, none of which are Pb.

Table 4: Top Important Features (v2)

Baseline v2: Top 10 Important Features	
log_pb_plus_sb	0.553805
pb_sb_over_non_ba_mass	0.416283
pb_ba_over_non_sb_mass	0.010844
pb_ba_over_non_sb_o_mass	0.005890
ba_sb_over_non_pb_mass	0.002172
ba	0.001582
sb	0.001485
as	0.000607
pb_sb_over_non_ba_o_mass	0.000584
ba_sb_over_non_pb_o_mass	0.000576

Table 5: Top Important Features (v3)

Baseline v3: Top 10 Important Features	
pb	0.761106
sb	0.105771
gsr_over_confounders	0.081863
ba	0.013410
pb_ba_over_non_sb_mass	0.007247
ca	0.006718
al	0.006696
ba_sb_over_non_pb_mass	0.003500
as	0.002371
cu	0.002097

In stark contrast, the right table shows how v3 feature importance has 7 out of 10 raw elements and only 3 engineered features. Lead (Pb) is absent from v2's top 10 but ranks 1st for v3 at 76%, with antimony (Sb) ranking 2nd at 10%. This strongly supports the idea that splits on lead and antimony gave v3 a fast start and then hit a ceiling as the remaining top features are all barium-related or confounder-related features.

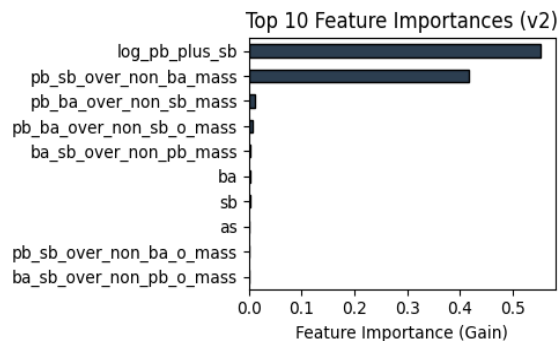


Figure 5: Top Feature Importances Bar (v2)

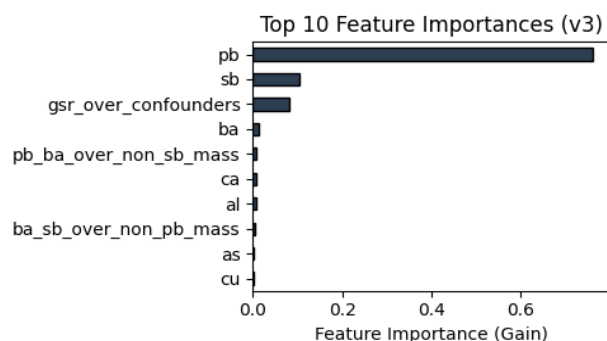


Figure 6: Top Feature Importances Bar (v3)

Overall, there appears to be slightly better feature distribution for v3, at least across the top 2 through 10. With optimized parameter tuning, the extractable learning signal from validation could potentially be smoothed out to enhance the models performance and reduce false positives.

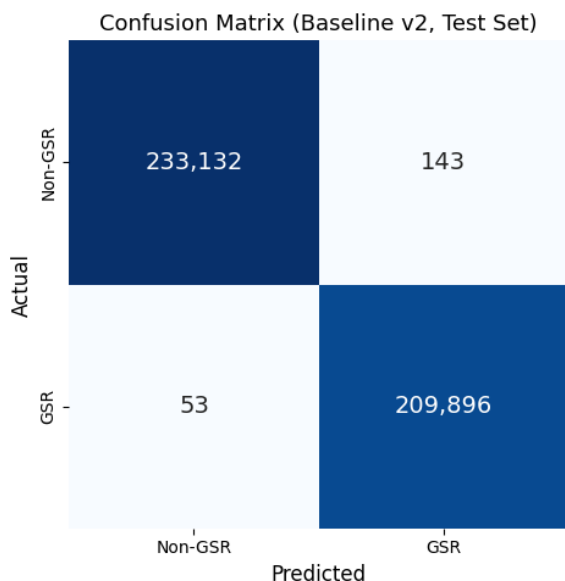


Figure 7: Confusion Matrix Heatmap (v2)

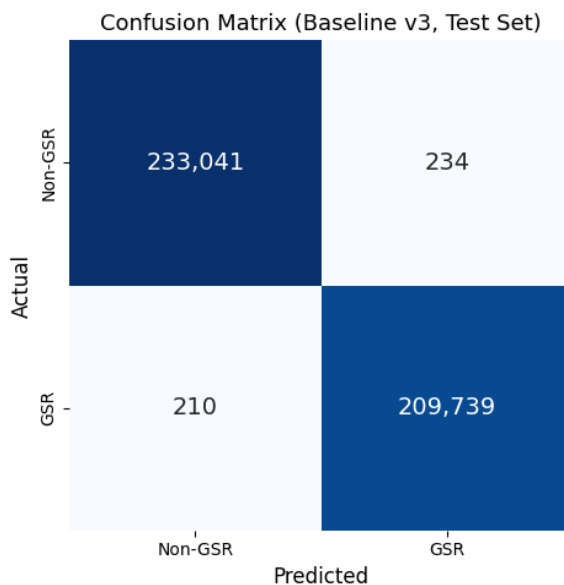


Figure 8: Confusion Matrix Heatmap (v3)

The confusion matrix heatmaps in the figures above illustrate just how similar these models are performing for actual vs. predicted. From a legal perspective, the key square to consider is the top-right (false positives), which can be summarized appropriately by calculating the false positive rate (FPR). For v2, the FPR is 0.06%, and for v3 it is 0.1%. For both model versions, the leading culprit for false positives is barium, specifically when teamed up with aluminum or in some cases calcium and silicon. The tables below show the distribution of false positives for each model version by particle class. For v2, barium-heavy particle classes are 96% of the false positives. For v3, they make up 99% of false positives.

Table 6: False Pos. by Particle Class (v2)

FP by class:	
final_class	
BaAl	100
BaCaSi	37
CuZn	3
TiZnGd	3

Table 7: False Pos. by Particle Class (v3)

FP by class:	
final_class	
BaAl	187
BaCaSi	44
TiZnGd	3

In terms of the XGBoost assumptions described in section 3.2, this model type is a non-parametric, tree-based ensemble method that doesn't assume linearity, normally distributed features, homoscedasticity, or feature independence. Traditional diagnostic tests such as Durbin-Watson or residual normality plots are inapplicable. Instead, assumptions can be addressed through heuristics and diagnostics appropriate for tree-based models. Not all assumptions are exclusively addressed by this report.

1. The assumption of label correctness was partially tested through edge case analysis during XGBoost's feature engineering, where 36 Non-GSR particles across the entire binary dataset contained all 3 core GSR elements (lead, antimony, and barium) and 5,092 Non-GSR particles had 2 out of 3 core GSR elements. Chemical profiling confirmed that the particles contained only trace amounts of lead or antimony, signalling that these were not mislabelled but rather contaminated samples.
2. The assumption of feature informativeness is supported by the feature importance distribution (Figures 5 & 6) as well as the overall performance of the baseline tree models. In some capacity, whether as raw elements or engineered features, both lead and antimony have proven to be incredibly informative, accounting for the vast majority of split importance.

3. As previously mentioned, the assumption that the training data is representative of the deployment distribution is supported by the stub-level splitting strategy that ensures generalization to unseen physical samples within the NFI dataset.
4. Assumptions of appropriate model complexity are addressed via hyperparameters, although for this exercise standard default values were selected for the baseline. Even still, these mechanisms seem validated as described by the paragraph on “overfitting” at the end of this section.

Assumptions 5 and 6 are timelessness and uniformity, which unfortunately cannot be directly addressed because we don’t have access to comprehensive metadata for collection methods, collection dates, quantification methods, etc. Nevertheless, these remain necessary assumptions for the scope of this research.

Overfitting diagnostics will be revisited in greater detail during the hyperparameter tuning phase, where we will systematically vary regularization parameters and evaluate their effect on the train-validation gap. For this report, there is no sign of classical overfitting. The learning curve from figures 3 and 4 show train and validation AUC-PR tracking closely with no divergence. In the case of figure 4 (v3), early stopping is evident as the boosting is terminated to prevent overfitting. Lastly, three-fold cross-validation was implemented using StratifiedGroupKFold for both v2 and v3 baseline models. The results of the combined train and validation sets had low variance across folds, yielding mean PR-AUC of 0.999998 ± 0.000001 with mean FPR of 0.000525 ± 0.000216 for v2, mean PR-AUC 0.999997 ± 0.000002 with mean FPR of 0.000533 ± 0.000231 for v3. The conclusion is that both models generalize stably across different stub groupings.

Table 8: Comparison of Eval Metrics between XGBoost baseline models (v2) and (v3)

Metric	Baseline v2	Baseline v3
Features	40	37
Accuracy	0.999912	0.998998
Precision (GSR)	0.999319	0.998886
Recall (GSR)	0.999748	0.998999
ROC-AUC	0.999997	0.99997
PR-AUC	0.999996	0.99995
False Positives	143	234
FPR	0.000613	0.001003
Early Stopping Round	208	53
CV Mean PR-AUC	0.999998 $\pm$ 0.000001	0.999997 $\pm$ 0.000002
CV Mean FPR	0.000525 $\pm$ 0.000216	0.000533 $\pm$ 0.000231

4.3 Neural Network (MLP)

The initial baseline run revealed near perfect performance, which can be an indicator of leakage. On the hold out test set at the 0.5 classification threshold, the model achieved 99.9% accuracy, ROC-AUC 1.00, and PR-AUC 1.00. The confusion matrix shows 233,122 true negatives, 209,823 true positives, 153 false positives, and 125 false negatives. A subclass analysis revealed that the classic three element GSR signature (PbBaSb) achieves perfect recall (1.00), while classes with the highest errors are BaAl (specificity 0.9986) and BaSb (recall 0.9954). Both of these particles contain Barium, which is consistent with EDA findings that Ba is the primary source of false positives due to its presence in both GSR and non GSR particles.

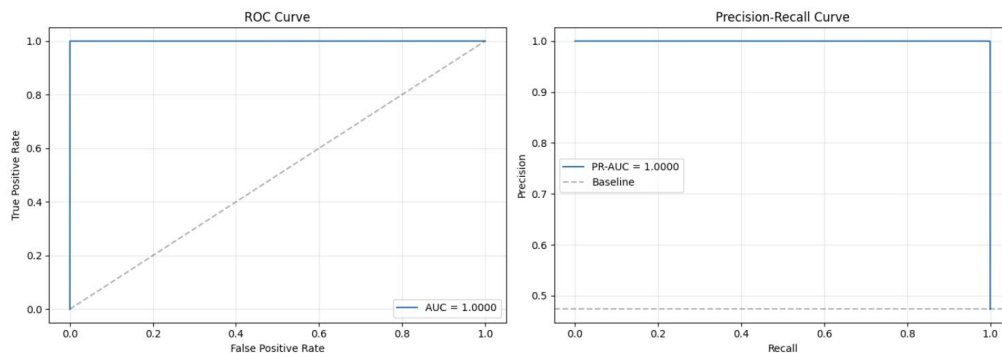


Figure 9: MLP ROC Curve and PR Curve

Upon review, we suspect that the GSR count feature indeed introduces a form of target leakage, as it describes the count of Pb, Ba, and Sb present in a particle sample. We concluded that this feature is a likely proxy for our label. In the second version of the baseline, we dropped the GSR count feature, reducing our feature space to 56 predictors. However, performance remained near perfect even without it. Our results indicate that a binary GSR vs non GSR classification is a trivial task when the data contains strong signal and well separated feature space, like in the multi element signatures (e.g. PbBaSb is GSR, CuZn is non GSR). The model correctly learns this rule. The more forensically meaningful test is whether the model handles particles that fall outside of this rule or descends into catastrophic failure (at random classifications).

In order to test this, we ran the model on the 506,682 ambiguous particles that were initially excluded from feature engineering. These are mostly single element classes, such as Sb (225,676 samples), Pb (168,334 samples), Ba (89,277 samples), and Sr (23,395 samples). These samples could represent either GSR fragments, or stand alone environmental confounders. See table X for detailed performance metrics.

Table 9: Model predictions on ambiguous particles

<i>Class</i>	<i>N</i>	<i>GSR %</i>	<i>Mean P(GSR)</i>	<i>Median P(GSR)</i>	<i>P(GSR) < 0.2</i>	<i>0.2 < P(GSR) < 0.8</i>	<i>P(GSR) > 0.8</i>
Ba	89,277	0.4%	0.004	0.00	99.5%	0.2%	0.3%
Pb	168,334	1.8%	0.018	0.00	98.2%	0.0%	1.8%
Sb	225,676	2.1%	0.02	0.00	97.8%	0.4%	1.8%
Sr	23,395	0.1%	0.001	0.00	99.8%	0.1%	0.0%

The model classified 97-99% of these particles as non GSR with high confidence ($P(\text{GSR}) < 0.2$). Only 1.8% of the Pb only and 2.1% of the Sb only particles received GSR predictions, while Ba only (0.4%) and Sr only (0.1%) scored negligible as GSR.

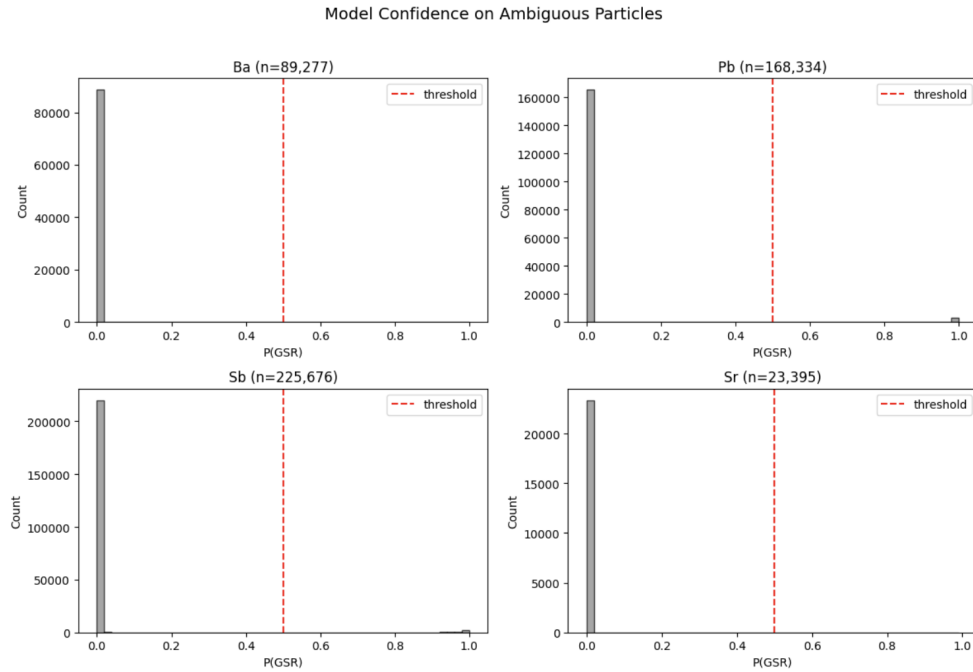


Figure 10: Model Confidence on Ambiguous Particles

The probability distributions for all particles are heavily centered around 0 with no substantial mass in the uncertain ranges. This result also aligns with forensic domain knowledge, where particles containing only lead do not match the classic GSR signature and therefore are more likely of environmental origin. The discriminative signal learned by the model is the cooccurrence of multiple GSR elements. The low confidence for the single element samples reflects the underlying aleatoric uncertainty where the ambiguity stems from the sample itself not the model limitations. This validates our original NFI labeling decision to classify these particles as ambiguous and supports the decision of deferring to expert judgements in single element samples where further analysis of spectra and additional context may be necessary.

It is noteworthy that the MLP is not overfitting, despite near perfect performance. First, the training and validation loss curves track closely through training epochs, with the

final gaps between train loss and val loss being small (0.0023 vs 0.0015). Second, the cross validation results produce consistent results with a mean ROC-AUC of 1.0 and mean accuracy of 0.9992 (+- 0.0001) and mean false positive rate of 0.0012 (+- 0.004). Low variance across folds indicates stable generalization properties. Lastly, we applied dropout regularization during training and early stopping on val loss to prevent overfitting. Overfitting is not a concern regarding near perfect performance, instead we ask the more relevant question of whether the model learned the decision boundary to generalize to out of distribution samples (e.g. from NIST), which will be tested in the next phase.

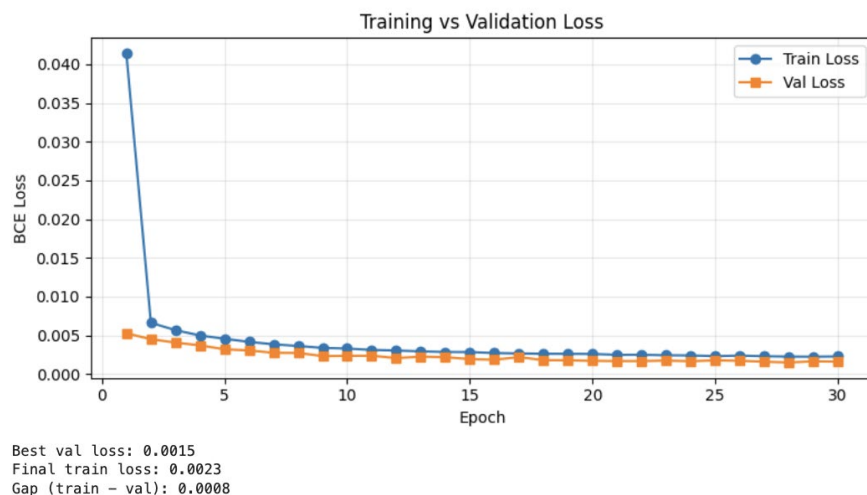


Figure 11: Train vs val loss over 30 epochs

5. Discussion and Next Steps

5.1 Logistic Regression

The logistic regression results confirm that a relatively simple linear model is sufficient to achieve strong classification performance on the binary GSR versus Non-GSR task. When dominant elemental signatures (Pb, Ba, Sb) are included, the classification problem becomes nearly trivial, with the model achieving near-perfect performance driven by a small number of highly predictive features.

However, this near-perfect performance highlights an important limitation: the model is heavily reliant on deterministic signals that effectively encode the target variable. While this behavior is expected given the known chemical composition of GSR particles, it reduces the model's value for understanding more complex or ambiguous classification scenarios.

The reduced model, which excludes Pb, Ba, and Sb, provides a more meaningful baseline by forcing the model to rely on broader compositional patterns. Although performance decreases to a ROC-AUC of approximately 0.90, the model remains effective, demonstrating that additional predictive structure exists beyond the primary GSR elements. This version of the model is therefore more representative of real-world conditions, where dominant signals may be absent or less clearly defined.

A key advantage of logistic regression is its interpretability. Coefficient analysis provides direct insight into how individual elemental features influence classification decisions, which is particularly important in a forensic context where transparency and explainability are critical for courtroom defensibility. The contrast between the original and reduced models further reinforces the importance of understanding feature dominance when interpreting model outputs.

Despite its strengths, logistic regression is limited in its ability to capture non-linear relationships and complex feature interactions. While it performs well for clearly separable classes, more flexible models such as gradient-boosted trees and neural networks may better capture subtle patterns in borderline or ambiguous particle classes.

As a next step, logistic regression will continue to serve as a benchmark model for comparison against more complex approaches. Future work will include evaluation on ambiguous particle classes, assessment of probability calibration, and validation on external datasets to better understand model generalizability and failure modes.

5.2 XGBoost

The baseline analysis for XGBoost revealed several key findings. First, the binary GSR classification task is highly separable, even when using raw elemental measurements with minimal engineered features. This addresses our primary research question. And in terms of which features drive predictions, lead and antimony concentrations provide sufficient signal for near-perfect classification when paired with minimal engineered features like lead & antimony ratio over confounders, or lead & barium ratio over total non-antimony mass, or barium & antimony ratio over total non-lead mass. The combination of these specific features has proven to have better distribution across the top 10 important features.

Secondly, the systematic identification and removal of feature leakage and/or label proxies as described in Appendix E, namely `core_gsr_count`, demonstrates that engineered features can easily be misinterpreted, introduce leakage, and cause a

model to overperform on test sets. These label proxies are more reflective of the expert labeling process than the chemical predictors and elemental compositions.

Third, false-positive analysis for the baseline tree models directly addresses one of our secondary research questions regarding the conditions for model failure. An overwhelming majority of false positive misclassifications belong to barium-heavy particle classes like BaAl and BaCaSi, accounting for 99% of the test set's false positives. This strongly supports the notion that barium-rich environmental or industrial particles are the leading cause of confusion in GSR classification.

The initial next step for the XGBoost model is selecting between v2 and v3. Some additional exploration can be done, perhaps by reincorporating the sparse elements that were dropped during pre-processing since the tree model should be able to handle the sparsity without issue. Evaluating performance against the ambiguous particle set may also help guide the decision. It is important to consider the very slight tradeoff between performance and interpretability, where v2 achieves a lower false-positive rate but is about 97% reliant on engineered features which may or may not be cause for concern in a legal setting when interpreting results. On the other hand, the baseline for v3 has a slightly higher false-positive rate but more interpretable results that rely mostly on raw element composition with support from very specific engineered features.

In either case, critical next steps also include hyperparameter tuning, which will play a key role in improving the model from the baseline by using RandomizedSearchCV with StratifiedGroupKFold and experimenting with adjustments like max_depth, learning_rate, n_estimators, regularization, and subsampling rates. Additionally, utilizing

the precision-recall curve to optimize the classification threshold where target precision is achieved at maximum recall, balanced with the cost of false positives. Lastly, SHAP analysis will be conducted using TreeExplainer for global feature importance, waterfall plots for additional false-positive analysis, and dependence plots for key feature interactions.

5.3 Neural Network

Our findings directly address our revised research question of whether machine learning models can reliably predict GSR vs non GSR particles when the multi element signatures are present. However, the forensically more interesting finding comes from the model behavior on ambiguous particles. The overwhelming majority of samples was classified as non GSR with high confidence, suggesting that the model has indeed learned a decision boundary consistent with forensic expert judgements. The discovery and removal of the GSR count feature was an important methodological lesson, highlighting the risk of feature engineering to introduce leakage.

Our original analysis plan called for a simple MLP baseline followed by hyperparameter tuning. Given that our baseline achieves near perfect performance, hyperparameter tuning is unlikely to give more lift in key KPIs however. We therefore revise our plan going forward to focus on the forensically harder and more valuable questions:

1. TabNet (Arik & Pfister 2021). We will implement TabNet as our primary neural network model, replacing our baseline MLP. TabNet uses sequential attention mechanisms to perform instance wise feature selection, producing interpretable attention masks that show which elements the model relies on for prediction.

Interpretability is one of our main goals, and ML models providing forensic experts with understandable outputs is critical for courtroom admissibility. We will use the PyTorch TabNet implementation from <https://github.com/dreamquark-ai/tabnet>.

2. MC Dropout Uncertainty Estimation. We will apply Monte Carlo dropout to quantify uncertainty around model predictions, particularly for the ambiguous particles. This is critical to understand what drives model failure modes. By running multiple stochastic forward passes with dropout at inference time, we can obtain a distribution of predictions per sample. The variance serves as our uncertainty estimate. The results can give us valuable insights into confident prediction cases vs uncertain ones, providing a principles analysis of flagging samples for expert review.
3. Threshold Optimization. The default 0.5 classification threshold may not be appropriate for our use case. False positives carry a significant cost in forensic applications. We will use precision-recall curves to identify thresholds that minimize false positives while maintaining acceptable recall, informed by the real life cost in the courtroom.

Citations

Citations

Arik, S. Ö., & Pfister, T. (2021, May). Tabnet: Attentive interpretable tabular learning. In *Proceedings of the AAAI conference on artificial intelligence* (Vol. 35, No. 8, pp. 6679-6687).

Matzen, T., Kukurin, C., van de Wetering, J., Ariëns, S., Bosma, W., Knijnenberg, A., ... & Ypma, R. J. (2022). Objectifying evidence evaluation for gunshot residue comparisons using machine learning on criminal case data. *Forensic science international*, 335, 111293.

Ritchie, Nicholas W. M., Renolds, Amy (2021), Automated particle analysis (SEM/EDS) data from samples known to have been exposed to gunshot residue and from samples occasionally mistaken for gunshot residue - like brake dust and fireworks., National Institute of Standards and Technology, <https://doi.org/10.18434/mds2-2476> (Version: 1.0, Accessed 2026-03-21)

Appendices

Appendix A: GitHub Repository (*Team Delta Capstone*)

<https://github.com/bkoconnell/datascience-capstone>

Source Data (engineered features):

- Logistic Regression: **data/processed/engineered_features_logistic.parquet**
- XGBoost: **data/processed/engineered_features_xgboost.parquet**
- Neural Network: **data/processed/engineered_features_nn.parquet**

Baseline Model Notebooks → **notebooks/03_model_exploration**

Appendix B: Data Dictionary XGBoost

https://github.com/bkoconnell/datascience-capstone/tree/main/artifacts/reports/03_baseline_model/data_dictionaries/xgb_model_dataadict.md

Appendix C: Data Dictionary Neural Network

The neural network feature dataset (engineered_features_nn.parquet) contains 2,294,985 rows and 62 columns.

Metadata Columns

Column	Type	Description
stub_id	int	SEM stub identifier
particle_id	int	Particle identifier within stub
label	string	"GSR" or "Non_GSR"
target	int	Binary target: GSR = 1, Non_GSR = 0
final_class	string	Particle subclass (e.g., PbBaSb, CuZn, BaAl)

CLR-Transformed Element Features (27 columns)

Each column is the Centered Log-Ratio transform of the corresponding element's weight-percent concentration. Values are real-valued and unconstrained (can be negative). Naming convention: *clr_{element}* .

Column	Element
clr_o	Oxygen
clr_cl	Chlorine
clr_cr	Chromium
clr_s	Sulfur
clr_k	Potassium
clr_br	Bromine

clr_cu	Copper
clr_na	Sodium
clr_mo	Molybdenum
clr_ba	Barium
clr_mg	Magnesium
clr_sr	Strontium
clr_al	Aluminum
clr_ti	Titanium
clr_ni	Nickel
clr_si	Silicon
clr_sn	Tin
clr_w	Tungsten
clr_ca	Calcium
clr_p	Phosphorus
clr_hg	Mercury
clr_pb	Lead
clr_mn	Manganese
clr_sb	Antimony
clr_as	Arsenic
clr_fe	Iron

Binary Presence Features (27 columns)

Each column is 1 if the corresponding element has concentration > 0 , and 0 otherwise.

Naming convention: *has_{element}* .

Engineered Features (3 columns)

Column	Formula	Description
pb_sb_over_non_ba_o_mass	$(\text{Pb} + \text{Sb}) / (\text{total_mass} - \text{Ba} - \text{O})$	Mass fraction of Pb+Sb excluding Ba and O; sentinel value -1 when denominator = 0
log_pb_plus_sb	$\log_{10}(\text{Pb} + \text{Sb})$	Log-transformed additive GSR signal
gsr_count	count of {Pb, Ba, Sb} > 0	Number of core GSR elements present (0–3)

Appendix D: Data Dictionary Logistic Regression

Metadata Columns

Column	Type	Description
stub_id	int	SEM stub identifier
particle_id	int	Particle identifier within stub
label	string	“GSR”, “Non_GSR”, or “Ambiguous”
target	int	Binary target: GSR = 1, Non_GSR = 0

Log-Transformed Element Features

Column	Description
log_pb	Log-transformed lead concentration
log_ba	Log-transformed barium concentration
log_sb	Log-transformed antimony concentration

Ratio Features

Column	Formula	Description
log_pb_sb_ratio	$\log(\text{Pb} / \text{Sb})$	Relative relationship between Pb and Sb
log_pb_ba_ratio	$\log(\text{Pb} / \text{Ba})$	Relative relationship between Pb and Ba
log_ba_sb_ratio	$\log(\text{Ba} / \text{Sb})$	Relative relationship between Ba and Sb

GSR vs Confounder Features

Column	Formula	Description
log_pb_zn_ratio	$\log(\text{Pb} / \text{Zn})$	Distinguishes GSR from environmental particles
log_sb_cu_ratio	$\log(\text{Sb} / \text{Cu})$	Separates GSR from copper-rich particles
log_ba_ti_ratio	$\log(\text{Ba} / \text{Ti})$	Differentiates industrial/environmental sources
log_gsr_over_confounders	Combined ratio	Overall GSR signal vs confounders

Composition Features

Column	Description
pb_pct	% of particle mass from Pb
ba_pct	% of particle mass from Ba
sb_pct	% of particle mass from Sb

Interaction Features

Column	Description
pb_sb_interaction	Interaction between Pb and Sb
pb_ba_interaction	Interaction between Pb and Ba
ba_sb_interaction	Interaction between Ba and Sb

Appendix E: Feature Leakage / Label Proxies

https://github.com/bkoconnell/datascience-capstone/tree/main/artifacts/reports/03_baseline_model/appendix_label_proxy/xgboost_feature_leakage_eval.md